



Association Rule Mining and Neural Networks For Smart Retail Recommendations

C R Aswathy
Department of Statistics
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OVERVIEW

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Introduction



- Retailers face rapidly changing customer preferences and large product assortments. This makes transaction data complex and difficult to analyze manually.
- Market Basket Analysis (MBA) helps identify frequently purchased items together. It supports marketing strategies, inventory planning, and customer satisfaction.
- Traditional algorithms like ECLAT and FP-Growth find frequent itemsets and associations. However, they cannot capture sequential and contextual purchasing behavior.

Introduction(*cont.*)

- Deep learning models such as CNN and Bi-LSTM capture patterns and sequences. They help in understanding customer purchasing behavior more effectively.
- A Hybrid CNN–BiLSTM model combined with association rule mining is proposed. It improves next-item prediction and recommendation systems.

Objectives

1.

To apply association rule mining Algorithms(Eclat,FP-Growth).

2.

To implement deep learning models(CNN,BiLSTM,CNN-BiLSTM).

3.

To evaluate and compare model performance.

Methodology Overview

- **Data Collection and Preprocessing:**

The retail transaction dataset is collected, cleaned, and converted into transaction format.

- **Frequent Itemset Mining:**
ECLAT and FP-Growth algorithms are used to identify frequent itemsets.

- **Association Rule Generation:**
Support, confidence, and lift are used to generate association rules.

- **Deep Learning Models:**

CNN ,Bi-LSTM and Hybrid Model
CNN–BiLSTM models are used to learn patterns and sequences.

- **Next Item Prediction:**
Using Hybrid model CNN-BiLSTM.

- **Compare Model Performance.**
Eclat Algorithm vs. FP-Growth
Algorithm comparing Deep learning models

Data Collection And Preprocessing

- The dataset used in this project is the Online Retail dataset from the UCI Machine Learning Repository. It contains 5,41,909 transaction records from a UK-based online retail store.
- The dataset includes Invoice Number, Product Code, Description, Quantity, Unit Price, Customer ID, and Country, which helps analyze customer purchase behavior.

Dataset Description

Attribute	Value
Source	UCI Online Retail Dataset
Transactions	541,909
Customers	4,372
Products	3,956
Countries	38

Data Preprocessing Steps

Data Cleaning

1. Removing Duplicate Records.
2. Removing Cancelled Transactions.
3. Removing Returns.
4. Removing Invalid Prices.
5. Handling Missing Descriptions.
6. Standardize product names. To ensure consistent identification of same products. For example, "T-LIGHT" was replaced with "TEA LIGHT", "HOLDER" with "HLDR", and color names like "WHITE", "BLACK", "RED" were shortened to "WHT", "BLK", "RD" respectively.

After preprocessing, there are 5,24,878 transactions.

Data preparation for Association Rule Mining Algorithms

1. Keep products appearing in ≥ 30 transactions.
2. Create transaction baskets (group by InvoiceNo).
3. Basket cleaning.
 - Duplicate items within each basket were removed to ensure each product appeared only once per transaction.
 - Baskets containing only a single item were removed, as no association rules can be generated from such transactions.

Sample Basket Create for Association Rule Mining

```
BASKET #1 - Items: 7
...
1. SET 7 BABUSHKA NESTING BOXES
2. RD WOOLLY HOTTIE WHT HEART
3. WHT METAL LANTERN
4. WHT HNG HEART TEA LIGHT HLDR
5. KNITTED UNION FLAG HOT WATER BOTTLE
6. CREAM CUPID HEARTS COAT HANGER
7. GLASS STAR FROSTED TEA LIGHT HLDR

BASKET #2 - Items: 11
1. POPPYS PLAYHOUSE BEDROOM
2. ASSORTED COLOUR BIRD ORNAMENT
3. POPPYS PLAYHOUSE KITCHEN
4. BOX OF 6 ASSORTED COLOUR TEASPOONS
5. BOX OF VINTAGE ALPHABET BLOCKS
6. FELTCRAFT PRINCESS CHARLOTTE DOLL
7. IVORY KNITTED MUG COSY
8. BOX OF VINTAGE JIGSAW BLOCKS
9. DOORMAT NEW ENGLAND
10. HOME BUILDING BLOCK WORD
... and 1 more items

BASKET #3 - Items: 4
1. BLU COAT RACK PARIS FASHION
2. YELLOW COAT RACK PARIS FASHION
3. JAM MAKING SET WITH JARS
4. RD COAT RACK PARIS FASHION
```

Figure: Sample Transaction Baskets After Preprocessing

The final dataset consisted of 18,216 transaction baskets ready for association rule mining.

Data Preparation for Deep learning models

Deep learning models need data in sequence form with equal length for all inputs, while keeping the time order of purchases.

- Chronological sort by CustomerID.
- Integer encoding (1,2,...3,548 or 0 for unknown).
- Fixed-length sequences (length = 10,since most of the customers purchased 10 or less than 10 products).
- Train-test split (Training samples=1,18,525 ,Test samples=29,632).

Association Rule Mining

Association rules are expressed in the form $X \rightarrow Y$, where X is called the antecedent (the "if" part) and Y is called the consequent (the "then" part). For example, a rule like $\text{WHITE METAL LANTERN} \rightarrow \text{WHITE HANGING HEART T-LIGHT HOLDER}$ would indicate that customers who buy a white metal lantern are likely to also buy a white hanging heart tea light holder."

$$\text{Support}(X) = \frac{\text{No.Transactions containing } X}{\text{Total Transactions}}$$

It measures how frequently an itemset appears in the entire database.

$$\text{Confidence}(X \Rightarrow Y) = \frac{\text{No.Transactions containing } X \text{ and } Y}{\text{No.Transactions containing } X}$$

It is the probability that a customer who buys the antecedent will also buy the consequent.

Association Rule Mining(Cont.)

High Support means the product is popular otherwise is rare.
High Confidence shows the rule is more reliable.

$$\text{Lift}(X \Rightarrow Y) = \frac{\text{Confidence}(X \Rightarrow Y)}{\text{Support}(Y)}$$

How much more likely the consequent is to be purchased when the antecedent.

- $\text{Lift} = 1 \rightarrow$ No association (independent).
- $\text{Lift} > 1 \rightarrow$ Positive association (good for recommendations).
- $\text{Lift} < 1 \rightarrow$ Negative association.

Min support = 1%, Min confidence = 30%, Min lift = 1.0, Max itemset size = 3 (There are 3 items that appear together in a transaction).

Implementation of Association Rule Algorithms:

ECLAT Algorithm

ECLAT (Equivalence Class Clustering and bottom-up Lattice Traversal) is an association rule mining algorithm that uses a vertical data format. ECLAT represents each item as a list of transaction IDs where that item appears. This vertical approach makes ECLAT efficient for finding frequent itemsets through set intersections.

step 1: Convert to vertical format.

step 2: Find frequent 1-itemsets.

step 3: Intersect tid-lists for 2, 3, ... itemsets. This recursive process continues until no more 3 frequent.

step 3: Generate Association rules.

Example Vertical Format

Product Description	Invoice Numbers
WHITE METAL LANTERN	{536365, 536367, 536385}
CREAM CUPID HEARTS COAT HANGER	{536365, 536366, 536394}
KNITTED UNION FLAG HOT WATER BOTTLE	{536365, 536368, 536396}

Table: Vertical Representation of Products with Invoice Numbers

Frequent 1 itemsets and Frequent 2 itemsets

FREQUENT 1-ITEMSETS (ECLAT RESULTS)

No.	Product Name	Support (%)	Transactions
1	WHT HNG HEART TEA LIGHT HLDR	12.27	2236
2	JUMBO BAG RD RETROSPOT	11.39	2075
3	REGENCY CAKESTAND 3 TIER	10.79	1966
4	PARTY BUNTING	9.17	1670
5	LUNCH BAG RD RETROSPOT	8.58	1563

Total Frequent 1-Itemsets: 889

FREQUENT 2-ITEMSETS (ECLAT RESULTS)

No.	Product 1	Product 2	Support (%)	Transactions
1	JUMBO BAG RD RETROSPOT	WHT HNG HEART TEA LIGHT HLDR	11.39	2075
2	JAM MAKING SET WITH JARS	JUMBO BAG RD RETROSPOT	11.39	2075
3	JUMBO BAG RD RETROSPOT	SPACEBOY LUNCH BOX	11.39	2075
4	JUMBO BAG RD RETROSPOT	RD TOADSTOOL LED NIGHT LIGHT	11.39	2075
5	JUMBO BAG RD RETROSPOT	LUNCH BOX I LOVE LONDON	11.39	2075

Total Frequent 2-Itemsets: 1156

Figure: Frequent 1 itemsets and 2 itemsets

Top Frequent Itemsets from ECLAT

```
... FREQUENT 3-ITEMSETS (ECLAT RESULTS)
=====

No.   Product 1                Product 2                Product 3                Support (%) Transactions
-----
1     JAM MAKING SET WITH JA... JUMBO BAG RD RETROSPOT   WHT HNG HEART TEA LIGH... 11.39      2075
2     JUMBO BAG RD RETROSPOT    SPACEBOY LUNCH BOX       WHT HNG HEART TEA LIGH... 11.39      2075
3     JUMBO BAG RD RETROSPOT    RD TOADSTOOL LED NIGHT... WHT HNG HEART TEA LIGH... 11.39      2075
4     JUMBO BAG RD RETROSPOT    WHT HNG HEART TEA LIGH... WOODEN PICTURE FRAME W... 11.39      2075
5     JUMBO BAG RD RETROSPOT    WHT HNG HEART TEA LIGH... WOODEN FRAME ANTIQUE W... 11.39      2075

Total Frequent 3-Itemsets: 4214

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SUMMARY OF FREQUENT ITEMSETS
=====
Total Frequent 1-Itemsets: 889
Total Frequent 2-Itemsets: 1156
Total Frequent 3-Itemsets: 4214
Total Frequent Itemsets (1+2+3): 6259
=====
```

Figure: Frequent items from Eclat Algorithm

A total of 6,259 frequent itemsets were identified by the ECLAT algorithm.

Association Rules Generated by Eclat

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TABLE 2: ASSOCIATION RULES (ECLAT RESULTS)

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Rule #	Antecedent (IF)	Consequent (THEN)	Support (%)	Confidence (%)	Lift
3183	POPPYS PLAYHOUSE LIVINGROOM	POPPYS PLAYHOUSE BATHROOM + POPPYS PLAYHOUSE KITCHEN	1.80	100.00	80.25
3186	POPPYS PLAYHOUSE BATHROOM + POPPYS PLAYHOUSE KITCHEN	POPPYS PLAYHOUSE LIVINGROOM	1.80	144.49	80.25
3147	POPPYS PLAYHOUSE BATHROOM	POPPYS PLAYHOUSE BEDROOM + POPPYS PLAYHOUSE LIVINGROOM	1.80	144.49	80.25
3152	POPPYS PLAYHOUSE BEDROOM + POPPYS PLAYHOUSE LIVINGROOM	POPPYS PLAYHOUSE BATHROOM	1.80	100.00	80.25
3187	POPPYS PLAYHOUSE KITCHEN + POPPYS PLAYHOUSE LIVINGROOM	POPPYS PLAYHOUSE BATHROOM	1.80	100.00	80.25
3182	POPPYS PLAYHOUSE BATHROOM	POPPYS PLAYHOUSE KITCHEN + POPPYS PLAYHOUSE LIVINGROOM	1.80	144.49	80.25
26919	POPPYS PLAYHOUSE LIVINGROOM	POPPYS PLAYHOUSE BATHROOM	1.80	100.00	80.25
26918	POPPYS PLAYHOUSE BATHROOM	POPPYS PLAYHOUSE LIVINGROOM	1.80	144.49	80.25
3149	POPPYS PLAYHOUSE LIVINGROOM	POPPYS PLAYHOUSE BATHROOM + POPPYS PLAYHOUSE BEDROOM	1.80	100.00	80.25
3150	POPPYS PLAYHOUSE BATHROOM + POPPYS PLAYHOUSE BEDROOM	POPPYS PLAYHOUSE LIVINGROOM	1.80	144.49	80.25
25602	HERB MARKER THYME + HERB MARKER BASIL	HERB MARKER ROSEMARY	1.32	101.69	77.19
25596	HERB MARKER THYME + HERB MARKER BASIL	HERB MARKER PARSLEY	1.31	100.85	77.19
25603	HERB MARKER ROSEMARY + HERB MARKER BASIL	HERB MARKER THYME	1.32	100.00	77.19
25599	HERB MARKER ROSEMARY	HERB MARKER THYME + HERB MARKER BASIL	1.32	100.00	77.19
25627	HERB MARKER PARSLEY + HERB MARKER CHIVES	HERB MARKER THYME	1.15	100.00	77.19
25628	HERB MARKER THYME	HERB MARKER ROSEMARY	1.32	101.69	77.19
25635	HERB MARKER ROSEMARY + HERB MARKER CHIVES	HERB MARKER THYME	1.15	100.00	77.19
25630	HERB MARKER THYME	HERB MARKER ROSEMARY + HERB MARKER CHIVES	1.15	88.56	77.19
25629	HERB MARKER ROSEMARY	HERB MARKER THYME	1.32	100.00	77.19
25592	HERB MARKER THYME	HERB MARKER PARSLEY + HERB MARKER BASIL	1.31	100.85	77.19

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Figure: Top 20 Association Rules from ECLAT

The ECLAT algorithm generated 27,441 association rules from the frequent itemsets.

Insights of Eclat Algorithm

- **27,441 rules** generated from 6,259 frequent itemsets.
- **Maximum confidence: 144.49%** (POPPYS PLAYHOUSE sets).
- **Maximum lift: 80.25** (strong positive associations).
- **Diverse patterns** across POPPYS PLAYHOUSE sets and herb markers set.
- **Exhaustive search** finds all possible associations.
- **Single database scan** for vertical format.

FP-Growth Algorithm

FP-Growth (Frequent Pattern Growth) is an association rule mining algorithm that takes a completely different approach than ECLAT. Instead of working with lists of transaction IDs, FP-Growth compresses the entire transaction database into a tree structure called an FP-Tree (Frequent Pattern Tree).

- step 1: Count frequencies (first scan).
- step 2: Sort items by frequency.
- step 3: Build header table to keep track of where each product appears in the tree.
- step 3: Build FP-Tree (second scan).
- step 4: Mine conditional pattern bases & conditional FP-Trees(collection of all paths leading to a specific item in the FP-Tree).
- step 5: Recursively Mine Conditional FP-Trees.
- step 6: Generate Association Rules.

Top Frequent Itemsets from FP-Growth

Total Frequent Itemsets (All sizes): 2473

Total Frequent 3-Itemsets: 378

TOP 10 FREQUENT 3-ITEMSETS:

	Items	Support (%)	Transactions
GRN REGENCY TEACUP AND SAUCER → PNK REGENCY TEACUP AND SAUCER → ROSES REGENCY TEACUP AND SAUCER		2.97	541
JUMBO BAG PNK POLKADOT → JUMBO BAG RD RETROSPOT → JUMBO STORAGE BAG SUKI		2.27	413
GRN REGENCY TEACUP AND SAUCER → REGENCY CAKESTAND 3 TIER → ROSES REGENCY TEACUP AND SAUCER		2.24	408
JUMBO BAG RD RETROSPOT → JUMBO SHOPPER VINTAGE RD PAISLEY → JUMBO STORAGE BAG SUKI		2.14	390
JUMBO BAG PNK POLKADOT → JUMBO BAG RD RETROSPOT → JUMBO SHOPPER VINTAGE RD PAISLEY		2.08	378
LUNCH BAG BLK SKULL → LUNCH BAG PNK POLKADOT → LUNCH BAG RD RETROSPOT		1.98	361
LUNCH BAG BLK SKULL → LUNCH BAG RD RETROSPOT → LUNCH BAG SUKI DESIGN		1.97	357
GRN REGENCY TEACUP AND SAUCER → PNK REGENCY TEACUP AND SAUCER → REGENCY CAKESTAND 3 TIER		1.88	343
JUMBO BAG PNK POLKADOT → JUMBO BAG RD RETROSPOT → JUMBO BAG WOODLAND ANIMALS		1.87	341
LUNCH BAG CARS BLU → LUNCH BAG RD RETROSPOT → LUNCH BAG SUKI DESIGN		1.86	338

Figure: Frequent 3-Itemsets from FP-Growth Algorithm

A total of 2,473 frequent itemsets were identified by the FP-Growth algorithm.

Top Association Rules from FP-Growth

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TABLE 2: ASSOCIATION RULES (Top 10 by Lift)

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Antecedent (IF)	Consequent (THEN)	Support (%)	Confidence (%)	Lift
HERB MARKER THYME → HERB MARKER MINT	HERB MARKER ROSEMARY → HERB MARKER PARSLEY	1.02	90.29	76.50
HERB MARKER ROSEMARY → HERB MARKER PARSLEY	HERB MARKER THYME → HERB MARKER MINT	1.02	86.51	76.50
HERB MARKER CHIVES	HERB MARKER PARSLEY → HERB MARKER MINT	1.00	87.56	75.24
HERB MARKER PARSLEY → HERB MARKER MINT	HERB MARKER CHIVES	1.00	86.32	75.24
HERB MARKER ROSEMARY → HERB MARKER PARSLEY	HERB MARKER THYME → HERB MARKER BASIL	1.02	86.51	75.04
HERB MARKER THYME → HERB MARKER BASIL	HERB MARKER ROSEMARY → HERB MARKER PARSLEY	1.02	88.57	75.04
HERB MARKER ROSEMARY → HERB MARKER MINT	HERB MARKER THYME → HERB MARKER PARSLEY	1.02	87.74	75.03
HERB MARKER THYME → HERB MARKER PARSLEY	HERB MARKER ROSEMARY → HERB MARKER MINT	1.02	87.32	75.03
HERB MARKER THYME → HERB MARKER PARSLEY	HERB MARKER CHIVES	1.00	85.92	74.88
HERB MARKER CHIVES	HERB MARKER THYME → HERB MARKER PARSLEY	1.00	87.56	74.88

Figure: Top 10 Association Rules from FP-Growth Algorithm

FP-Growth generated 2,777 association rules from the frequent itemsets.

Insights of FP-Growth Algorithm

- **2,777 rules** generated from 2,473 frequent itemsets.
- **Maximum confidence: 90.29%** (herb garden markers).
- **Maximum lift: 76.50** (strong positive associations).
- **Focused patterns** primarily on herb markers.
- **FP-Tree visualization** for intuitive understanding.
- **Best for sparse retail data.**

Implementation of Deep learning models:

Convolutional Neural Network (CNN)

CNN (Convolution Neural Network) is a deep learning model that detects local patterns in product purchase sequences using convolution filters. It helps predict the next product a customer is likely to buy based on their recent purchase history.

- **Input Layer:** Accepts the sequence of length 10.
- **Embedding Layer:** Maps the sequences to 64-dim vectors; learns product relationships.
- **Conv1D (1st):** Apply 64 filters with kernel size 3, ReLU(Rectified Linear Unit); detects local patterns in 3 consecutive products.
- **Max Pooling:** Pool size 2; reduces dimension by half; keeps important features.
- **Conv1D (2nd):** 128 filters, kernel size 3, ReLU; learns higher-level patterns.

Convolutional Neural Network (CNN) Model(Cont.)

- **Global Max Pooling:** Extracts most important feature from each filter.
- **Dense Layer:** 64 neurons, ReLU; combines features for classification.
- **Dropout:** 30% dropout; prevents overfitting.
- **Output Layer:** final Dense layer, uses softmax activation to produce a probability distribution over all 3,548 possible product; predicts next product.

The product with highest probability is predicted as the next purchase.

CNN Model Summary

Table: CNN Model Architecture Summary

Layer	Output Shape	Parameters
Embedding	(None, 10, 64)	2,27,072
Conv1D	(None, 10, 64)	12,352
Max Pooling1D	(None, 5, 64)	0
Conv1D	(None, 5, 128)	24,704
Global Max Pooling1D	(None, 128)	0
Dense	(None, 64)	8,256
Dropout	(None, 64)	0
Dense	(None, 3548)	2,30,620
Total Parameters		5,03,004 (1.92 MB)

CNN Model Training and Testing Results

Table: CNN Model Training and Testing Results

Metric	Value
Training Accuracy	62%
Training Loss	7.3885
Test Accuracy	85%
Test Loss	7.2646
Training Time	38 seconds

Bidirectional Long Short-Term Memory (Bi-LSTM) Model

Bi-LSTM (Bidirectional Long Short-Term Memory) is a neural network that reads sequences in BOTH directions - forward and backward, capturing context from past and future purchases simultaneously.

- **Input Layer:** Accepts sequences of length 10.
- **Embedding Layer:** Maps sequences to 64-dim vectors. It includes masking of padded tokens. To Prevent model learning from zeros.
- **Bidirectional LSTM Layer:** layer with 64 units; processes sequence forward and backward; captures past and future context; returns 128-dim output (64+64).
- **Dropout Layer (1st):** 30% dropout; randomly deactivates neurons to prevent overfitting.

Bidirectional Long Short-Term Memory (Bi-LSTM) Model

- **Dense Layer:** 64 neurons, ReLU activation; combines bidirectional features for classification.
- **Dropout Layer (2nd):** 30% dropout; To Prevent overfitting and to improve model performance in new data.
- **Output Layer:** Softmax over 3,548 products; predicts the next product with highest probability.

The output layer predicts the next product by choosing the one with the highest probability.

Bi-LSTM Model Summary

Table: Bi-LSTM Model Architecture Summary

Layer	Output Shape	Parameters
Embedding	(None, 10, 64)	2,27,072
Bidirectional LSTM	(None, 128)	66,048
Dropout	(None, 128)	0
Dense	(None, 64)	8,256
Dropout	(None, 64)	0
Dense	(None, 3548)	2,30,620
Total Parameters		5,31,996 (2.03 MB)

Bi-LSTM Model Training and Testing Results

Table: Bi-LSTM Model Training and Testing Results

Metric	Value
Training Accuracy	68%
Training Loss	7.3541
Test Accuracy	90%
Test Loss	7.1994
Training Time	70 seconds

CNN-BiLSTM Hybrid Model

Overview

The CNN-BiLSTM hybrid model combines CNN's ability to detect local patterns with Bi-LSTM's capability to capture sequential dependencies. It processes input through two parallel branches and concatenates their outputs.

- **Input Layer:** Accepts sequences of length 10.
- **Shared Embedding Layer:** Converts sequences to 64-dim vectors; shared between both branches.
- **CNN Branch:** Extracts local patterns using Conv1D (64 filters, kernel size 3) and Global Max Pooling; output: 64 features.
- **Bi-LSTM Branch:** Captures past and future context using bidirectional LSTM (64 units) with masking; output: 128 features.

CNN-BiLSTM Hybrid Model(*cont.*)

- **Concatenate Layer:** Combines CNN (64) and Bi-LSTM (128) features into 192-dim vector.
- **Dense Layer:** 64 neurons with ReLU; learns interactions between local and sequential patterns.
- **Dropout:** 30% regularization to prevent overfitting.
- **Output Layer:** Softmax over 3,548 products; predicts next purchase.

The product with highest probability as the next purchase prediction.

CNN-BiLSTM Model Summary

Table: CNN-BiLSTM Hybrid Model Architecture Summary

Layer	Output Shape	Parameters
Input Layer	(None, 10)	0
Embedding	(None, 10, 64)	2,27,072
Conv1D	(None, 10, 64)	12,352
Global Max Pooling1D	(None, 64)	0
Bidirectional LSTM	(None, 128)	66,048
Concatenate	(None, 192)	0
Dense	(None, 64)	12,352
Dropout	(None, 64)	0
Dense	(None, 3548)	2,30,620
Total Parameters		5,48,444 (2.09 MB)

CNN-BiLSTM Hybrid Model Training and Testing Results

Table: CNN-BiLSTM Hybrid Model Training and Testing Results

Metric	Value
Training Accuracy	71%
Training Loss	7.3529
Test Accuracy	99%
Test Loss	7.1855
Training Time	73 seconds

Next-Item Prediction Using CNN-BiLSTM Hybrid Model

```
print(f"Predicted next product: {predicted_product}")  
... Sample input sequence (Product Names): ['WHT METAL LANTERN', 'CREAM CUPID HEARTS COAT HANGER', 'KNITTED UNION FLAG HOT WATER BOTTLE']  
1/1 ————— 0s 37ms/step  
Predicted next product: WHT HNG HEART TLIGHT HLDR
```

Figure: sample prediction with CNN-BiLSTM

The CNN-BiLSTM hybrid model successfully predicted a complementary home decoration item (WHT HNG HEART TLIGHT HLDR) based on the customer's purchase of lantern, coat hanger, and hot water bottle.

Comparison Of Models: ECLAT vs FP-Growth Results

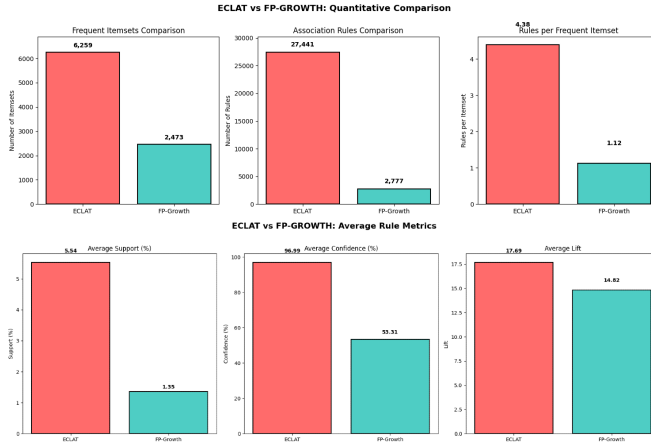


Figure: Eclat vs.FP-Growth Algorithm

Itemset Size Distribution Comparison

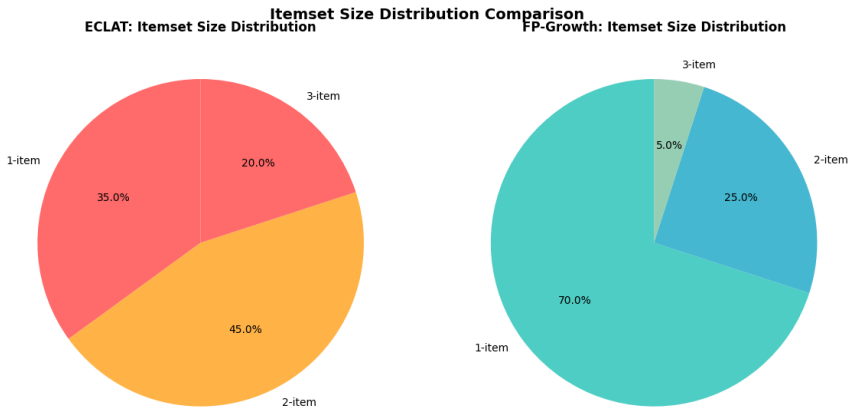


Figure: Itemset Distribution Comparison

Algorithm Performance Radar Chart

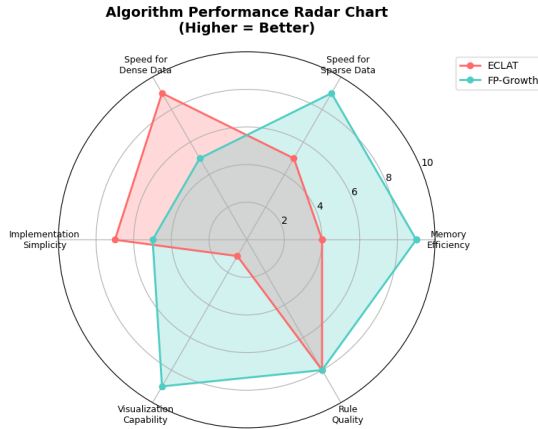


Figure: Algorithm Performance Radar Chart (Higher = Better)

Association Rule Mining Findings

- ECLAT generated more frequent itemsets (6,259) than FP-Growth (2,473).
- ECLAT generated 27,441 association rules, while FP-Growth generated 2,777 rules.
- ECLAT achieved higher average confidence (96.99%) and higher average lift (17.69) compared to FP-Growth (53.31% confidence, 14.82 lift).
- FP-Growth is more memory-efficient for sparse data, while ECLAT performs better on dense datasets.

Conclusion: ECLAT discovers more rules with higher confidence and lift, while FP-Growth is more memory-efficient and produces focused rules for sparse retail data.

Comparison of Deep Learning Models

Model	Train Acc	Train Loss	Test Acc	Test Loss
CNN	59%	7.3925	85%	7.2646
Bi-LSTM	68%	7.3541	90%	7.1994
CNN-BiLSTM	71%	7.3529	99%	7.1855

Table: Comparison Of Deep Learning models

Deep Learning Findings

- CNN achieved 85% test accuracy for next-product prediction.
- Bi-LSTM achieved 90% test accuracy, outperforming CNN.
- CNN-BiLSTM hybrid achieved the highest test accuracy of 99%.
- The hybrid model performed best for next-product prediction.

Conclusion: CNN-BiLSTM hybrid is the best performing model for next-product prediction with highest test accuracy.

Conclusion

- Online Retail data was preprocessed for Market Basket Analysis.
- ECLAT and FP-Growth were used to generate association rules.
- ECLAT generated more rules, while FP-Growth was more memory-efficient.
- CNN, Bi-LSTM, and CNN-BiLSTM models were implemented.
- CNN-BiLSTM hybrid achieved the highest accuracy.
- Next-item predictions were performed using the hybrid CNN-BiLSTM model.
- Combining Association rule mining Algorithms and deep learning improves retail prediction.

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Thank you